

ENUNCIADO — PROBLEMA 1

Math 685: Preliminary Exam – Fall 2013

AUGUST 21

Instructions: This is a closed book, closed notes in class exam. You are required to do any four out of five problems but are strongly encouraged to attempt all five of them.

1. (a) *Projections:* Let H be a linear space with scalar product $\langle \cdot, \cdot \rangle$ and corresponding norm $\|\cdot\|$. Let S be a subspace of H , and let $u \in H$. Prove that $u^* \in S$ is the best approximation of u within S if and only if u^* is the orthogonal projection of u onto S that is

$$\langle u - u^*, v \rangle = 0 \quad \forall v \in S.$$

Hint: argue with the function $f(t) = \|u - (u^* + tv)\|^2$.

- (b) *Bessel Inequality:* Let $u \in L^2(a, b) := \left\{ f : \int_a^b |f(x)|^2 dx < \infty \right\}$. Let $\{p_i\}_{i=1}^n$ be a set of orthonormal functions in $L^2(a, b)$ and $S := \text{span} \{p_i\}_{i=1}^n$. Let $u_n = \sum_{i=1}^n \alpha_i p_i \in S$ be the best (least squares) approximation of u . Show that $\|u - u_n\|^2 = \|u\|^2 - \|u_n\|^2$ and conclude

$$\sum_{i=1}^n |\alpha_i|^2 \leq \|u\|^2.$$

2. *Gaussian Quadrature.* Derive the two point Gaussian integration formula $G_2(f)$ for

$$I(f) = \int_a^b f(x) dx.$$

Indicate for what polynomial degree this rule is exact.

3. Let $b \in \mathbb{R}^n$ be given and $A \in \mathbb{R}^{n \times n}$ be a strictly diagonally dominant matrix, namely,

$$\max_{1 \leq i \leq n} \sum_{j \neq i} \left| \frac{a_{ij}}{a_{ii}} \right| = r < 1.$$

Show that the *Jacobi* iteration $x_{k+1} = D^{-1} (L + U) x_k + D^{-1} b$ converges regardless of the initial guess $x_0 \in \mathbb{R}^n$.

Hint: Recall, $A = D - L - U$, where D is diagonal, L is lower triangular and U is upper triangular.

4. The following MATLAB script can be used to compute its eigenvalues of $A \in \mathbb{R}^{n \times n}$:

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1 B = A;
2 for k = 1:m
3     [Q, R] = qr(B);
4     B      = RQ;
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- (a) Show that consecutive iterates B are similar, and deduce they are all similar to A (Recall that two square matrices B, C are similar if $C = SBS^{-1}$).

SOLUCIÓN — PROBLEMA 1

Problema 1 — Proyecciones y desigualdad de Bessel

(a) Teorema de mejor aproximación / proyección ortogonal.

($\Rightarrow$) Supongamos $u^* = \arg \min_{v \in S} \|u - v\|$. Para cualquier $v \in S$ y $t \in \mathbb{R}$, $u^* + tv \in S$, así que

$$f(t) = \|u - (u^* + tv)\|^2 = \|u - u^*\|^2 - 2t\langle u - u^*, v \rangle + t^2\|v\|^2 \geq f(0).$$

El mínimo en $t = 0$ implica $f'(0) = 0$, es decir $-2\langle u - u^*, v \rangle = 0$ para todo $v \in S$: $\langle u - u^*, v \rangle = 0$ para todo $v \in S$. ■

($\Leftarrow$) Si $\langle u - u^*, v \rangle = 0$ para todo $v \in S$, entonces para cualquier $w \in S$:

$$\|u - w\|^2 = \|u - u^* + u^* - w\|^2 = \|u - u^*\|^2 + \|u^* - w\|^2 + 2\langle u - u^*, u^* - w \rangle.$$

Como $u^* - w \in S$, el producto cruzado es 0, y $\|u^* - w\|^2 \geq 0$, así $\|u - w\|^2 \geq \|u - u^*\|^2$ con igualdad solo si $w = u^*$: u^* es la mejor aproximación. ■

(b) Desigualdad de Bessel. Sea $u_n = \sum_{i=1}^n \alpha_i p_i$ la mejor aproximación en $S = \text{span}\{p_i\}$. Por (a), $\langle u - u_n, v \rangle = 0$ para todo $v \in S$, en particular para $v = u_n$:

$$\langle u - u_n, u_n \rangle = 0 \implies \langle u, u_n \rangle = \|u_n\|^2.$$

Entonces:

$$\|u - u_n\|^2 = \langle u - u_n, u - u_n \rangle = \langle u, u \rangle - 2\langle u, u_n \rangle + \|u_n\|^2 = \|u\|^2 - 2\|u_n\|^2 + \|u_n\|^2 = \|u\|^2 - \|u_n\|^2.$$

Como $\|u - u_n\|^2 \geq 0$: $\|u_n\|^2 \leq \|u\|^2$.

Los coeficientes óptimos son $\alpha_i = \langle u, p_i \rangle$ (proyección sobre la base ortonormal). Por la identidad de Parseval parcial:

$$\|u_n\|^2 = \sum_{i=1}^n \alpha_i^2 \langle p_i, p_i \rangle = \sum_{i=1}^n |\alpha_i|^2,$$

luego $\sum_{i=1}^n |\alpha_i|^2 \leq \|u\|^2$ para todo n . ■

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$$I(f) = \int_a^b f(x) dx.$$

Indicate for what polynomial degree this rule is exact.

3. Let $b \in \mathbb{R}^n$ be given and $A \in \mathbb{R}^{n \times n}$ be a strictly diagonally dominant matrix, namely,

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Problema 2 — Cuadratura gaussiana de 2 puntos

Buscamos la fórmula $G_2(f) = w_1 f(x_1) + w_2 f(x_2)$ exacta para polinomios de grado máximo posible. Con 4 parámetros (w_1, x_1, w_2, x_2) , se puede conseguir exactitud para $f = 1, x, x^2, x^3$ (grado $\leq 3 = 2 \cdot 2 - 1$, el teorema de Gauss garantiza esto eligiendo nodos = raíces del polinomio de Legendre P_2).

Nodos: $P_2(x) = \frac{1}{2}(3x^2 - 1)$ sobre $[-1, 1]$; raíces $x_{1,2} = \pm 1/\sqrt{3}$.

Pesos por integración exacta de $f = 1$: $w_1 + w_2 = \int_a^b dx$. Para $[a, b]$ general con cambio de variable $x = \frac{a+b}{2} + \frac{b-a}{2}t$, $t \in [-1, 1]$, los nodos son $x_i = \frac{a+b}{2} + \frac{b-a}{2}t_i$ con $t_i = \pm 1/\sqrt{3}$ y los pesos son $w_i = \frac{b-a}{2}$.

Para $[a, b]$: $G_2(f) = \frac{b-a}{2} \left[f\left(\frac{a+b}{2} - \frac{b-a}{2\sqrt{3}}\right) + f\left(\frac{a+b}{2} + \frac{b-a}{2\sqrt{3}}\right) \right]$.

Verificación sobre $[-1, 1]$ ($a = -1, b = 1$): $w_1 = w_2 = 1$, $x_{1,2} = \pm 1/\sqrt{3}$.

- $f = 1$: $1 + 1 = 2 = \int_{-1}^1 1 \, dx$. ✓
- $f = x$: $-1/\sqrt{3} + 1/\sqrt{3} = 0 = \int_{-1}^1 x \, dx$. ✓
- $f = x^2$: $1/3 + 1/3 = 2/3 = \int_{-1}^1 x^2 \, dx$. ✓
- $f = x^3$: $-1/(3\sqrt{3}) + 1/(3\sqrt{3}) = 0 = \int_{-1}^1 x^3 \, dx$. ✓
- $f = x^4$: $1/9 + 1/9 = 2/9 \neq 2/5 = \int_{-1}^1 x^4 \, dx$. ✗ (exacta solo hasta grado 3)

La fórmula es **exacta para polinomios de grado < 3** . ■

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Problema 3 — Convergencia de Jacobi para matrices estrictamente diagonalmente dominantes por filas

$A \in \mathbb{R}^{n \times n}$, $\max_{1 \leq i \leq n} \sum_{j \neq i} |a_{ij}| / |a_{ii}| = r < 1$.

La iteración de Jacobi: $G_J = -D^{-1}(L + U)$, con $(G_J)_{ij} = -a_{ij}/a_{ii}$ para $j \neq i$ y 0 en la diagonal.

Para la norma infinito: $\|G_J\|_\infty = \max_i \sum_{j \neq i} |-a_{ij}/a_{ii}| = \max_i \sum_{j \neq i} |a_{ij}|/|a_{ii}| = r < 1$.

Como $\rho(G_J) \leq \|G_J\|_\infty = r < 1$, la iteración converge. Más directamente:

Demostración directa: El error $e^{(k+1)} = G_J e^{(k)}$, así $\|e^{(k+1)}\|_\infty = \|G_J e^{(k)}\|_\infty \leq \|G_J\|_\infty \|e^{(k)}\|_\infty \leq r \|e^{(k)}\|_\infty$.

Por inducción: $\|e^{(k)}\|_\infty \leq r^k \|e^{(0)}\|_\infty \rightarrow 0$ para todo $x^{(0)}$, dado que $r < 1$. ■

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- (b) Show that A and B have the same eigenvalues (do not use any theorems but rather prove this fact directly).
- (c) Are the eigenvectors of A and B the same? If not, find a relation among them.
5. Consider the second order elliptic partial differential equation

$$-\operatorname{div}(\alpha(x)\operatorname{grad} u) = f \quad (1)$$

posed on a smooth domain $\Omega \subset \mathbb{R}^2$, subject to Dirichlet boundary conditions $u = 0$ on $\partial\Omega$.

- (a) Define $H_0^1(\Omega)$, and state a weak formulation of the problem (1). This should have the form

$$\text{find } u \in V \text{ such that } a(u, v) = (f, v) \text{ for all } v \in V, \quad (2)$$

where $a(\cdot, \cdot)$ and $(\cdot, \cdot)$ are bilinear forms and $V = H_0^1(\Omega)$.

- (b) Identify sufficient conditions on α that guarantee that

$$a(u, u) \geq \gamma \|u\|_V^2, \quad a(u, v) \leq \Gamma \|u\|_V \|v\|_V$$

for all $u, v \in V$, where $\|\cdot\|_V$ is a suitable norm defined on V .

- (c) Let V_h denote a finite element subspace of V . Define the discrete weak solution of (1) on V_h and show that if u is the solution to (2), then there exists a positive constant $C(\gamma, \Gamma)$ such that

$$\|u - u_h\|_V \leq C(\gamma, \Gamma) \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (3)$$

SOLUCIÓN — PROBLEMA 4

Problema 4 — Algoritmo QR: similitud, valores propios, vectores propios

(a) Un paso: $B_k = Q_k R_k$ (factorización QR de B_{k-1} , con $B_0 = A$), luego $B_{k+1} = R_k Q_k = Q_k^T B_k Q_k$ (porque $R_k = Q_k^T B_k$). Así $B_{k+1} = Q_k^T B_k Q_k$: B_{k+1} es similar a B_k (con matriz de similitud Q_k , ortogonal). Por inducción, todos los B_k son similares entre sí y por tanto todos similares a $B_0 = A$. ■

(b) B_k es similar a A : existe invertible S_k tal que $B_k = S_k A S_k^{-1}$. Si λ es valor propio de A ($Av = \lambda v$, $v \neq 0$), entonces $B_k(S_k v) = S_k A S_k^{-1} S_k v = S_k(Av) = \lambda(S_k v)$: λ es también valor propio de B_k . El argumento es simétrico, luego A y B_k tienen los mismos valores propios para todo k . ■

(c) En general los vectores propios de A y B_k **no coinciden**. Si v es vector propio de A ($Av = \lambda v$), entonces $S_k v$ es vector propio de B_k con el mismo λ . La relación es: vectores propios de $B_k = S_k \times$ (vectores propios de A). En la práctica, el producto $S_k = Q_1 Q_2 \cdots Q_k$ converge a la matriz de vectores propios de A .

ENUNCIADO — PROBLEMA 5

- (b) Show that A and B have the same eigenvalues (do not use any theorems but rather prove this fact directly).
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- (b) Identify sufficient conditions on α that guarantee that

$$a(u, u) \geq \gamma \|u\|_V^2, \quad a(u, v) \leq \Gamma \|u\|_V \|v\|_V$$

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Problema 5 — FEM para EDP elíptica

(a) $H_0^1(\Omega) = \{u \in L^2(\Omega) : |\nabla u| \in L^2(\Omega), u|_{\partial\Omega} = 0\}$ (espacio de Sobolev con traza cero).

Multiplicando $-\operatorname{div}(\alpha \nabla u) = f$ por $v \in H_0^1(\Omega)$ e integrando por partes:

$$a(u, v) := \int_{\Omega} \alpha \nabla u \cdot \nabla v \, dx = (f, v) := \int_{\Omega} f v \, dx \quad \forall v \in H_0^1(\Omega).$$

Formulación débil: encontrar $u \in V = H_0^1(\Omega)$ tal que $a(u, v) = (f, v)$ para todo $v \in V$.

(b) Condiciones en α :

- **Elipticidad (coercividad):** $\alpha(x) \geq \gamma_0 > 0$ en $\bar{\Omega}$ implica

$a(u, u) = \int_{\Omega} \alpha |\nabla u|^2 \geq \gamma_0 \|\nabla u\|_{L^2}^2 = \gamma_0 \|u\|_V^2$ (con $\|u\|_V = \|\nabla u\|_{L^2}$, norma en H_0^1 por desigualdad de Poincaré).

- **Continuidad (acotamiento):** $\alpha(x) \leq \Gamma_0 < \infty$ implica

$$|a(u, v)| \leq \Gamma_0 \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} = \Gamma_0 \|u\|_V \|v\|_V.$$

(c) Solución discreta: $u_h \in V_h \subset V$ tal que $a(u_h, v_h) = (f, v_h)$ para todo $v_h \in V_h$. **Lema de Galerkin (ortogonalidad):** $a(u - u_h, v_h) = 0$ para todo $v_h \in V_h$.

Estimación de Cea: para cualquier $v_h \in V_h$:

$$\gamma \|u - u_h\|_V^2 \leq a(u - u_h, u - u_h) = a(u - u_h, u - v_h) + a(u - u_h, v_h - u_h).$$

El segundo término es $a(u - u_h, v_h - u_h) = 0$ por ortogonalidad de Galerkin. Por continuidad del primero:

$$\gamma \|u - u_h\|_V^2 \leq \Gamma \|u - u_h\|_V \|u - v_h\|_V.$$

Dividiendo: $\|u - u_h\|_V \leq \frac{\Gamma}{\gamma} \|u - v_h\|_V$ para todo $v_h \in V_h$, luego

$$\|u - u_h\|_V \leq \frac{\Gamma}{\gamma} \inf_{v_h \in V_h} \|u - v_h\|_V, \quad C(\gamma, \Gamma) = \frac{\Gamma}{\gamma}. \quad \blacksquare$$